Dynamic Functional Connectivity in fMRI

COL786: ADVANCED FUNCTIONAL BRAIN IMAGING
Final Project Presentation

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Problem Statement

What is Functional Connectivity?

Definition

- In neuroscience, functional connectivity (FC) refers to the degree of correlation between time series data of two different Regions of Interest (ROI).
- If the BOLD time series of two regions go up and down together, we infer that those regions are functionally connected.

Frameworks for Computing FC

- **Static FC**: computes a single, time-averaged correlation over the entire duration of a scan, effectively assuming the brain’s network is in a stationary equilibrium.
- **Dynamic FC**: captures a non-equilibrium view by computing the correlation matrix for small windows, tracking how FC patterns evolve in time.

The Stationarity Problem

What is Stationarity?

- A stochastic process is **weakly stationary** if its underlying statistical properties (like mean and cross-covariance) remain constant over time.
- In fMRI, if the brain is stationary, true functional connectivity never changes; it is in a continuous equilibrium.

The Finite-Sample Trap

- **Why dFC isn't enough**: An observed dFC time series fluctuating over time is, on its own, NOT evidence that the brain is non-stationary. It could simply be finite-sample noise.
- Any random process (even a perfectly stationary one) will produce fluctuating correlations when chopped into short sliding windows.
- To prove the brain is genuinely changing states, we must test our dFC data against mathematically rigorous **null models**.

The Problem Statement

The Research Question

- Is the fMRI BOLD signal genuinely a non-stationary process, or is the observed variability merely a statistical byproduct of finite-sample noise?
- Can we mathematically quantify this (lack of) stationarity rather than relying on visual interpretation?

Our Approach

How We Approach It

- **1. End-to-End dFC Pipeline:** We apply sliding-window dynamic FC and k-means clustering to naturalistic fMRI data (N=30) to establish the baseline dynamics.
- **2. Surrogate Null Modeling:** We evaluate the real data against two mathematically rigorous surrogate-data null models (Phase Randomization and MVN-Gaussian) to strictly test for stationarity.

Methodology

Dataset & Brain Parcellation

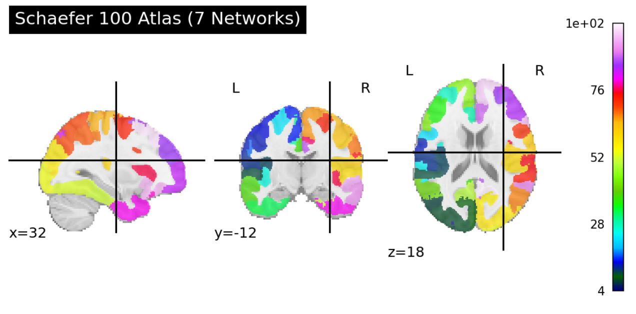
The Dataset

- **development_fmri** (via nilearn): N = 30 subjects watching a short Pixar movie inside the MRI scanner.
- TR = 2 seconds, 168 timepoints per subject.
- **Why naturalistic stimulus?** Subjects stay alert and engaged (unlike rest), without being constrained to a rigid task design.

Brain Parcellation

- Raw fMRI has $\sim$100,000 voxels, far too many for connectivity analysis.
- We use the **Schaefer 2018 atlas**: 100 cortical regions, organized into the **7 canonical Yeo networks** (Visual, Somatomotor, Dorsal Attn, Salience/VAN, Limbic, Frontoparietal, Default Mode).
- Average BOLD within each region gives one time series per ROI.

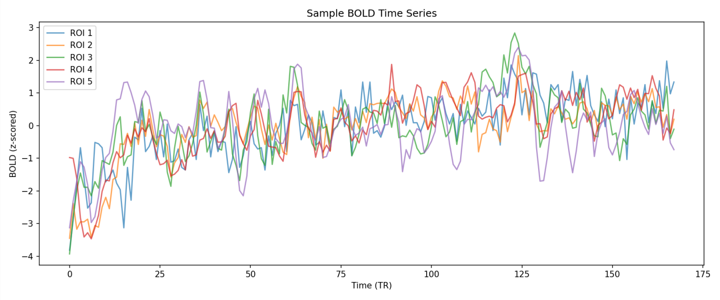
Dataset & Brain Parcellation



Preprocessing

Preprocessing Pipeline

- **Detrending**: removes slow scanner drift.
- **Bandpass filter (0.01–0.1 Hz)**: retains the canonical BOLD frequency band; removes very slow drift and high-frequency physiological noise.
- **Z-scoring**: standardizes each ROI to mean 0, variance 1, so correlations are interpretable.



Static Functional Connectivity

Computing FC

- For each subject, compute the Pearson correlation between every pair of ROIs:

$$\Sigma_{ij}^{(n)} = \frac{1}{T-1} \sum_{t=1}^{T} x_{t,i}^{(n)} x_{t,j}^{(n)}$$

- Yields a 100×100 symmetric matrix per subject.

Group Averaging via Fisher-z

- Correlations are bounded in $[-1, 1]$ — direct averaging is biased.
- Using $\tanh^{-1}$ to map to $(-\infty, \infty)$, average, then $\tanh$ back:

$$\bar{\Sigma}_{ij} = \tanh\left(\frac{1}{N} \sum_{n=1}^{N} \tanh^{-1}\left(\Sigma_{ij}^{(n)} \right) \right) \tag{1}$$

Visualizing the Static FC

What this shows

- Block-diagonal structure: ROIs in the same network are strongly correlated with each other.
- DMN and Frontoparietal networks show notable cross-coupling.
- This replicates the canonical 7-network organization of the human cortex.

Why this matters

- Static FC is our **baseline**. Anything dynamic FC reveals must be additional structure beyond this.

From Static to Dynamic FC

Sliding-Window Estimator

- For each subject, we slide a window of length L across time and compute a Pearson correlation matrix within each window:

$$\hat{\Sigma}^{(n)}(\tau) = \frac{1}{L-1} \tilde{X}^{(n,\tau)\top} \tilde{X}^{(n,\tau)}$$

Our Setup

- **Window length** $L = 30$ TRs (≈ 60 s)
- **Step size** $s = 1$
- **Number of windows per subject:** $W = T - L + 1 = 139$.
- **Hamming taper** smoothly down-weights edges of each window:

$$w[k] = 0.54 - 0.46 \cos\left(\frac{2\pi k}{L-1}\right)$$

Reduces spectral leakage from the sharp edges of a rectangular window.

Finding Recurring Patterns: Brain 'States'

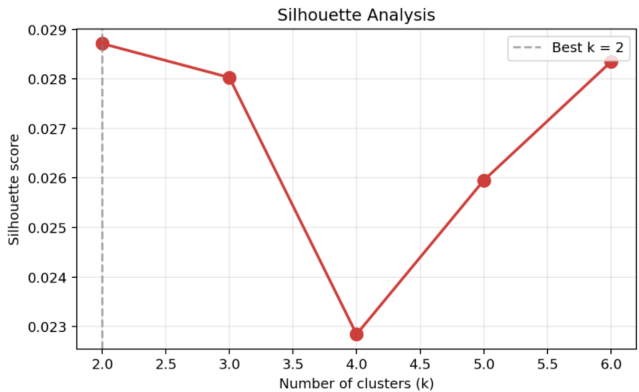
Motivation

- We have $30 \times 139 = 4{,}170$ windowed FC matrices.
- It is possible certain configurations recur across subjects and time.
- ***k*-means clustering** finds groups of similar FC matrices. Each cluster is a brain 'state'.

The Allen et al. (2014) Pipeline

1. Vectorize each FC matrix's upper triangle: $100 \times 100 \to \mathbb{R}^{4950}$.
2. Stack across all subjects and windows: $V \in \mathbb{R}^{4170 \times 4950}$.
3. Run k-means for $k \in \{2, \dots, 6\}$
4. Choose k via silhouette analysis.

Choosing the Number of States



- Silhouette score peaks at $k = 2$ (score $= 0.029$).
- **All values are extremely low** (≤ 0.029, theoretical max $= 1$): the data does not contain sharply separated clusters.

Why Sliding Windows Alone Are Not Enough

What This Implies

- Variability in dFC is **NOT the same** as non-stationarity.
- A perfectly stationary process, when chopped into short windows, will also produce fluctuating correlations due to finite-sample noise.
- **We cannot conclude non-stationarity from windowed FC alone.**
- We need a rigorous statistical test against **stationary null models**.

Surrogate Null Models

The Surrogate Testing Framework

The Big Idea

- Generate **artificial datasets** that are guaranteed to be stationary by construction.
- Run the *exact same* dFC pipeline on them.
- Compare a summary statistic between real and surrogate data.

The Decision Rule

- If the real data shows **significantly more** dFC variability than surrogates $\Rightarrow$ evidence for true non-stationarity.
- If the real data is **indistinguishable** from surrogates $\Rightarrow$ observed dynamics are explained by stationary noise.

Two Surrogate Models

- **Phase Randomization** — preserves power spectrum + static FC.
- **Multivariate Gaussian** — preserves only static FC.

Surrogate #1: Phase Randomization

Procedure (Prichard & Theiler, 1994)

1. Compute Fourier transform of each ROI: $\hat{x}_i(\omega) = \mathcal{F}[x_i(t)]$.
2. Generate **one** random phase shift $\phi(\omega) \sim U(0, 2\pi)$, **shared across all ROIs**.
3. Apply: $\tilde{x}_i(\omega) = \hat{x}_i(\omega) \cdot e^{i\phi(\omega)}$.
4. Inverse FFT: $\tilde{x}_i(t) = \mathcal{F}^{-1}[\tilde{x}_i(\omega)]$.

What This Preserves

- The **power spectrum** of every ROI (slow drifts intact).
- The **static cross-correlation** structure, because the same phase shift is applied to all ROIs at each frequency, cross-spectra (and hence static FC) are preserved.

What This Destroys

- Any non-linear or time-varying coupling between ROIs.

Surrogate #2: Multivariate Gaussian (MVN)

Procedure

1. Compute static covariance: $\Sigma = \mathrm{Cov}(X)$.
2. Cholesky factorization: $\Sigma = LL^\top$.
3. Sample i.i.d. white noise: $z(t) \sim \mathcal{N}(0, I)$.
4. Mix to get surrogate: $\tilde{x}(t) = Lz(t)$.

What This Preserves

- ONLY the static covariance matrix.

What This Destroys

- Power spectrum, autocorrelations, all temporal structure. Each timepoint is i.i.d.

The Variance Test

Summary Statistic: Total dFC Variance

- For each of the 4,950 unique edges, compute variance across 139 windows: V_{ij}.
- Sum across edges:

$$V_{\text{total}} = \sum_{i<j} \mathrm{Var}_w \left[\hat{\Sigma}_{ij}^{(w)} \right]$$

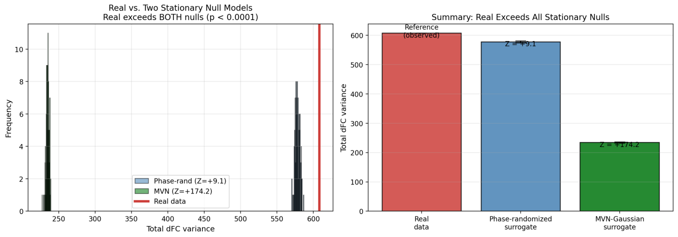
- One number per dataset that quantifies how much dFC fluctuates.

Procedure

- Compute $V_{\text{total}}^{\text{real}}$ on real data.
- Generate 100 surrogates per subject (per type), compute $V_{\text{total}}^{\text{surr}}$ for each.
- Compute Z-score against the surrogate distribution:

$$Z = \frac{V_{\text{total}}^{\text{real}} - \mu_{\text{surr}}}{\sigma_{\text{surr}}}$$

Result: Real vs Both Stationary Nulls



The Numbers

- Real total variance: ≈ 608
- Phase-rand: 580 ± 3
  $\Rightarrow Z = +9.1$
- MVN: 234 ± 2
  $\Rightarrow Z = +174.2$
- Both $p < 0.0001$.

Interpretation

- Real exceeds **both** nulls by a wide margin.
- Conclusion is robust to choice of null model.
- **The fMRI BOLD signal is genuinely non-stationary.**

Conclusions

Project Findings

What We Established

- The fMRI BOLD signal during naturalistic viewing is **conclusively non-stationary**.
- Verified by two independent surrogate null models:
  - vs Phase-randomized: $Z = +9.1$, $p < 0.0001$
  - vs MVN-Gaussian: $Z = +174.2$, $p < 0.0001$

Applications in Neuroscience

Clinical biomarkers

- Aberrant dFC patterns are reported in **schizophrenia** (Du et al. 2018), **major depressive disorder** (Zhi et al. 2018), and **Alzheimer's disease**.
- For these to be valid biomarkers, the underlying signal must be provably non-stationary, which our pipeline now lets us verify.

Cognitive & developmental neuroscience

- Tracking how connectivity reconfigures during **naturalistic tasks** (movie watching, narratives) probes real-world cognition.
- Studies of **brain development** and **aging** use fractional occupancy and dwell times as markers.

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Thank you!

Questions?